

Sergi Valverde, PhD

Tenured Scientist (Científico Titular), Spanish National Research Council (CSIC)

Institute of Evolutionary Biology (IBE, CSIC–UPF), Barcelona, Spain.

Theoretical research on how interactions, structural constraints and multiscale organization shape evolutionary change.

Contact and identifiers

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Legal name: Sergio Valverde Castillo

Curriculum summary

Sergi Valverde is a CSIC Tenured Scientist at the Institute of Evolutionary Biology (CSIC–UPF), Barcelona, where he leads the Evolution of Networks Laboratory. His research combines network theory, statistical physics, mathematical and computational modelling, and empirical analysis to investigate the organization and evolutionary dynamics of complex systems. Across biological, cultural and technological systems, he studies how interactions, structural constraints and processes operating across scales shape persistence, innovation and evolutionary transitions.

His recent work addresses higher-order organization in biological and ecological systems, multiscale and punctuated evolution, cultural and technological change, and the spatial dynamics of innovation. He has led competitive national research projects and the CSIC node of an international research consortium, supervised doctoral and postdoctoral researchers, and maintains external research affiliations with the European Centre for Living Technology and the University of Tennessee's Center for the Dynamics of Social Complexity.

His broader academic activity includes university teaching, European research evaluation, scientific-society governance, editorial work, and the organization of international scientific meetings.

Citation metrics: [Google Scholar](#) — 8,745 citations; h-index 38 (checked 1 October 2026).

International affiliations

1. **External Fellow**, European Centre for Living Technology (ECLT), Ca' Foscari University of Venice. Venice, Italy. Web: [ECLT website](#)
2. **External Fellow**, Center for the Dynamics of Social Complexity (DySoC), College of Arts and Sciences, University of Tennessee, Knoxville. Knoxville, USA. Web: [DySoC directory](#)

Education

- Informatics Engineering, Universitat Politècnica de Catalunya (UPC), Barcelona, Spain (13 Jul 1999).

- PhD in Applied Physics, Universitat Politècnica de Catalunya (UPC), Barcelona, Spain (19 Jan 2006).

Languages: Spanish, Catalan, English

Experience and job history

- Científico Titular (CSIC), Institute of Evolutionary Biology (CSIC-UPF), Barcelona, Spain; leader of the Evolution of Networks Laboratory (2018–present).
- Visiting Professor, Universitat Pompeu Fabra (UPF), Barcelona, Spain (2008–2018).
- Postdoctoral Researcher, Université Paul Sabatier, Toulouse, France (2007–2009).
- Postdoctoral Researcher, Universitat Pompeu Fabra (UPF), Barcelona, Spain (2006–2007).
- PhD Student, Universitat Pompeu Fabra (UPF), Barcelona, Spain (2004–2006).
- Software Engineer, Ubisoft Entertainment (1999–2004).
- R&D Engineer, El Periódico de Catalunya (1996–1999).

Research stays abroad (≥ 1 month)

- Santa Fe Institute (SFI), Santa Fe, USA — Research collaboration (Sep–Nov 2006; Mar–May 2007). Total duration: ~5 months.

Research projects (national and international)

International and transnational projects (participant / consortium member)

DYNAMICALLY EVOLVING, LARGE-SCALE INFORMATION SYSTEMS (DELIS)

Funding: Complex Systems Initiative, 6th Framework Programme (EU), grant agreement 001907.

Participants: Universität Paderborn; University of London; Consorzio Università Industria; Université Paris XI; Politechnika Wroclawska; Univerzita Karlova v Praze; Universitat Pompeu Fabra; Universität Karlsruhe; Max-Planck-Gesellschaft zur Förderung der Wissenschaft.

Duration: 2004-01-01 to 2008-02-29.

EU contribution: €4,500,000.

Total project cost: €6,831,002.

Coordinator (PI): Friedhelm Meyer auf der Heide.

Researchers involved: 20.

EMBODIED AND COMMUNICATING AGENTS (ECAGENTS)

Funding: Complex Systems Initiative, 6th Framework Programme (EU), grant agreement 001940.

Participants: Consiglio Nazionale delle Ricerche (coordinator); Université Libre de Bruxelles; Collegium Budapest, Institute for Advanced Study; Universitat Pompeu Fabra; IT-Forskningsinstitutet Viktoria AB; Humboldt Universität; Università La Sapienza; University of Tokyo; École Polytechnique Fédérale de Lausanne; Sony France S.A.

Duration: 2004-01-01 to 2008-03-31.

EU contribution: €4,300,986.

Total project cost: €7,115,589.

Coordinator (PI): Stefano Nolfi.

Researchers involved: 20+.

PROGRAMMABLE ARTIFICIAL CELL EVOLUTION (PACE)

Funding: Complex Systems Initiative, 6th Framework Programme (EU), grant agreement 002035.

Participants: Ruhr-Universität Bochum; Ca' Foscari University of Venice; University of Copenhagen; Chalmers University of Technology; Universitat Pompeu Fabra; University of Zurich; Vilnius University Institute of Theoretical Physics and Astronomy; ProtoLife s.r.l.; University of Southern Denmark; Comune di Venezia; Dublin City University; ProtoStream GmbH; Reed College.

Duration: 2004-04-01 to 2008-06-30.

EU contribution: €6,605,000.

Total project cost: €7,928,932.

Coordinator (PI): John McCaskill (Ruhr-Universität Bochum).

Researchers involved: 20+.

Competitive projects (national/institutional)

ORIGINS OF INNOVATION IN TINKERED NETWORKS

Funding: James McDonnell Foundation.

Host institution: Universitat Pompeu Fabra.

Duration: 2006-01-01 to 2010-12-31.

Total funding: €460,000.

PI: Ricard V. Solé.

Researchers involved: 2.

SOCIAL INSECT NESTS AS 3D NETWORKS: MORPHOGENESIS AND STRUCTURAL DESIGNS (MESOMORPH)

Funding: Agence Nationale de la Recherche (ANR), Biologie Systémique programme, ANR-06-BYOS-0008.

Host institution: Université Paul Sabatier.

Duration: 2007-01-01 to 2009-12-31.

Total funding: €500,000.

PI: Guy Theraulaz.

Researchers involved: 8.

FIS2009-12365 – Computación. Replicación y rotura de simetría en sistemas protocelulares

Funding: Ministerio de Ciencia e Innovación.

Host institution: Universitat Pompeu Fabra.

Duration: 2010-01-01 to 2013-12-31.

Total funding: €89,540.

PI: Ricard V. Solé.

Researchers involved: 7.

Principal Investigator (PI) projects

FIS2013-44674-P – Detección de tecnologías emergentes en redes de innovación

Funding: Ministerio de Economía y Competitividad.

Host institution: Universitat Pompeu Fabra.

Duration: 2013-01-01 to 2016-12-31.

Total funding: €14,520.

PI: Sergi Valverde.

FIS2016-77447-R – Predicción de innovación tecnológica en redes de culturómica

Funding: Ministerio de Economía y Competitividad.

Host institution: Universitat Pompeu Fabra.

Duration: 2016-12-30 to 2019-12-01.

Total funding: €41,140.

PI: Sergi Valverde.

PID2020-117822GB-I00 – Estructura y evolución del conectoma tecnológico

Funding: Agencia Estatal de Investigación.

Host institution: Consejo Superior de Investigaciones Científicas (CSIC), Institut de Biologia Evolutiva (CSIC–UPF).

Duration: 2021-09-01 to 2024-08-31.

Total funding: €90,750.

PI: Sergi Valverde.

PID2024-162055NB-I00 – Phylogenetic Scaling and Evolutionary Transitions in Mammalian Intelligence

Funding: Ministerio de Ciencia, Innovación y Universidades / Agencia Estatal de Investigación, Proyectos de Generación de Conocimiento (2024 call), co-funded by ERDF, EU.

Host institution: CSIC, Institut de Biologia Evolutiva (CSIC–UPF).

Duration: 2025-09-01 to 2028-08-31.

Total funding: €37,500 (€30,000 direct costs; €7,500 indirect costs).

PI: Sergi Valverde.

Researchers involved: 8.

INVESTIGO grant (Ministerio de Ciencia, Innovación y Universidades / SEPE)

Call: Proyectos Investigo 2025: public subsidy to finance the 12-month hiring of a young researcher (Grado $\geq$ 300 ECTS / Máster Universitario) in support of research projects.

Host institution: Institut de Biologia Evolutiva (CSIC–UPF), Barcelona.

Role: Sergi Valverde, host research group leader (research-support hiring subsidy).

Contract code: INV25SEPE_BAR_IBE_M3_2.

Total funding: €38,556.

Year awarded: 2025 (contract starts in 2026).

International coordination/node leadership

BiodivRestore-424 / MPA4Sustainability – Enhancing the role of MPAs in restoring biodiversity while maintaining access to ecosystem services

Funding: BiodivRestore Transnational Cofund (Innovation Fund Denmark; ANR; FCT; Agencia Estatal de Investigación; Swedish Environmental Protection Agency); CSIC award PCI2022-132936, funded by MCIN/AEI and the European Union NextGenerationEU/PRTR.

Participants: Technical University of Denmark; CNRS; Madeira Whale Museum; CSIC; Centre de Recerca Matemàtica / Autonomous University of Barcelona; Swedish University of Agricultural Sciences.

Duration: 2021-04-01 to 2024-03-31 (consortium dates published by Biodiversa).

CSIC award duration: 3 years from 2022-03-14, as awarded by the Agencia Estatal de Investigación.

Total consortium grant: €1,110,675.

CSIC node funding: €107,668.72 (€93,625 direct costs; €14,043.72 indirect costs).

Overall PI: David Lusseau.

CSIC Node Lead/Principal Researcher (Spain): Sergi Valverde.

Researchers involved: 20+.

Research lines

- Biological organization, higher-order assembly, and persistence in ecological and evolutionary systems.
- Multiscale evolutionary dynamics, punctuated change, and evolutionary transitions.
- Cultural and technological evolution, innovation, and the emergence and turnover of complex artifacts.
- Spatial dynamics of innovation, including fractal organization, urban scaling, and spatial inequality.

Publications

2001

1. Solé, R. V., & Valverde, S. (2001). Information transfer and phase transitions in a model of Internet traffic. *Physica A: Statistical Mechanics and its Applications*, 289(3–4), 595–605.

2002

2. Solé, R. V., Ferrer-i-Cancho, R., Montoya, J. M., & Valverde, S. (2002). Selection, tinkering, and emergence in complex networks. *Complexity*, 8(1), 20–33.
3. Valverde, S., & Solé, R. V. (2002). Self-organized critical traffic in parallel computer networks. *Physica A: Statistical Mechanics and its Applications*, 312(3–4), 636–648.

4. Valverde, S., et al. (2002). Scale-free networks from optimal design. *Europhysics Letters*, 60(4), 512.

2003

5. Valverde, S., & Solé, R. V. (2003). Hierarchical small worlds in software architecture. arXiv, cond-mat/0307278.

2004

6. Buhl, C., et al. (2004). Efficiency and robustness in ant networks of galleries. *The European Physical Journal B: Condensed Matter and Complex Systems*, 42(1), 123–129.
7. Solé, R. V., & Valverde, S. (2004). Information theory of complex networks: On evolution and architectural constraints. In *Complex Networks* (pp. 189–207). Springer, Berlin, Heidelberg.
8. Valverde, S. (2004). A new science for a connected world. *Current Anthropology*, 45(4), 565–566.
9. Valverde, S., & Solé, R. V. (2004). Internet's critical path horizon. *The European Physical Journal B*, 38(2), 245–252.

2005

10. Valverde, S. (2005). Evolution and dynamics in information networks (PhD thesis). Department of Applied Physics, Technical University of Catalonia, Barcelona, Spain.
11. Valverde, S., & Solé, R. V. (2005). Logarithmic growth dynamics in software networks. *Europhysics Letters*, 72(5), 858.
12. Valverde, S. (2005). Network motifs in computational graphs: A case study in software architecture. *Physical Review E*, 72(2), 026107.

2006

13. Buhl, C., et al. (2006). Topological patterns in street networks of self-organized urban settlements. *The European Physical Journal B: Condensed Matter and Complex Systems*, 49(4), 513–522.
14. Solé, R. V., & Valverde, S. (2006). Are network motifs the spandrels of cellular complexity? *Trends in Ecology & Evolution*, 21(8), 419–422.
15. Valverde, S., & Solé, R. V. (2006). On the nature of design. In *Complex Engineered Systems: Science Meets Technology* (pp. 72–100). Springer, Berlin, Heidelberg.
16. Valverde, S., et al. (2006). Self-organization patterns in wasp and open-source communities. *IEEE Intelligent Systems*, 21(2), 36–40.

2007

17. Corominas-Murtra, B., et al. (2007). K-scaffold subgraphs of complex networks. *Europhysics Letters*, 77(1), 18004.
18. Corominas-Murtra, B., et al. (2007). Emergence of scale-free syntax networks. arXiv, 0709.4344.
19. Perna, A., et al. (2007). The topological fortress of termites. In *Workshop on Bio-inspired Design of Networks* (pp. 165–173). Springer, Berlin, Heidelberg.
20. Rosas-Casals, M., et al. (2007). Topological vulnerability of the European power grid under errors and attacks. *International Journal of Bifurcation and Chaos*, 17(7), 2465–2475.
21. Solé, R. V., et al. (2007). Modularity in biological networks. In *Biological Networks* (pp. 21–40).
22. Valverde, S. (2007). Crossover from endogenous to exogenous activity in open-source software development. *Europhysics Letters*, 77(2), 20002.

23. Valverde, S., & Solé, R. V. (2007). Self-organization versus hierarchy in open-source social networks. *Physical Review E*, 76(4), 046118.
24. Valverde, S., et al. (2007). Topology and evolution of technology innovation networks. *Physical Review E*, 76(5), 056118.

2008

25. Challet, D., & Valverde, S. (2008). Fat tails, long memory, maturity and ageing in open-source software projects. arXiv, 0802.3170.
26. Mestres, J., et al. (2008). Data completeness—the Achilles heel of drug-target networks. *Nature Biotechnology*, 26(9), 983–984.
27. Perna, A., et al. (2008). The structure of gallery networks in the nests of termite *Cubitermes* spp. revealed by X-ray tomography. *Naturwissenschaften*, 95(9), 877–884.
28. Perna, A., et al. (2008). Topological efficiency in three-dimensional gallery networks of termite nests. *Physica A: Statistical Mechanics and its Applications*, 387(24), 6235–6244.
29. Solé, R. V., & Valverde, S. (2008). Spontaneous emergence of modularity in cellular networks. *Journal of the Royal Society Interface*, 5(18), 129–133.
30. Solé, R. V., Macía, J., Fellermann, H., Munteanu, A., Sardanyés, J., & Valverde, S. (2008). Models of protocell replication. In S. Rasmussen, M. A. Bedau, L. Chen, D. Deamer, D. C. Krakauer, N. H. Packard, & P. F. Stadler (Eds.), *Protocells: Bridging Nonliving and Living Matter* (pp. 213–231). MIT Press. DOI: [10.7551/mitpress/9780262182683.003.0010](https://doi.org/10.7551/mitpress/9780262182683.003.0010).
31. Solé, R. V., et al. (2008). Robustness of the European power grids under intentional attack. *Physical Review E*, 77(2), 026102.

2009

32. Corominas-Murtra, B., et al. (2009). Emergence of scale-free syntax networks. In *Evolution of Communication and Language in Embodied Agents* (pp. 83–101). Springer, Berlin, Heidelberg.
33. Corominas-Murtra, B., et al. (2009). The ontogeny of scale-free syntax networks: Phase transitions in early language acquisition. *Advances in Complex Systems*, 12(3), 371–392.
34. Mestres, J., et al. (2009). The topology of drug–target interaction networks: Implicit dependence on drug properties and target families. *Molecular BioSystems*, 5(9), 1051–1057.
35. Valverde, S., & Solé, R. V. (2009). Motifs. In *Encyclopedia of Complexity and Systems Science* (Vol. 13, pp. 5692–5702). Springer.
36. Valverde, S., et al. (2009). Percolation in insect nest networks: Evidence for optimal wiring. *Physical Review E*, 79(6), 066106.

2010

37. Horta-Bernús, R., et al. (2010). Discerning electricity consumption patterns from urban allometric scaling. In *Complexity in Engineering* (pp. 49–51). IEEE.
38. Solé, R. V., et al. (2010). Language networks: Their structure, function, and evolution. *Complexity*, 15(6), 20–26.

2011

39. Flores, C. O., et al. (2011). Statistical structure of host–phage interactions. *Proceedings of the National Academy of Sciences*, 108(28), E288.

40. Solé, R. V., et al. (2011). Convergent evolutionary paths in biological and technological networks. *Evolution: Education and Outreach*, 4(3), 415–426.

2012

41. Valverde, S., & Solé, R. V. (2012). Motifs in graphs. In *Computational Complexity* (pp. 1919–1928). Springer, New York, NY.
42. Valverde, S., et al. (2012). Evolved modular epistasis in artificial organisms. *Artificial Life*, 13, 111–115.

2013

43. Flores, C. O., et al. (2013). Multi-scale structure and geographic drivers of crossinfection within marine bacteria and phages. *The ISME Journal*, 7(3), 520–532.
44. Flores, C. O., et al. (2013). Phage-bacteria infection networks: From nestedness to modularity. In *APS March Meeting Abstracts* (J44–011).
45. Solé, R. V., & Valverde, S. (2013). Before the endless forms: Embodied model of transition from single cells to aggregates to ecosystem engineering. *PLOS ONE*, 8(4), e59664.
46. Solé, R. V., & Valverde, S. (2013). Macroevolution in silico: Scales, constraints and universals. *Palaeontology*, 56(6), 1327–1340.
47. Solé, R. V., et al. (2013). The evolutionary ecology of technological innovations. *Complexity*, 18(4), 15–27.
48. Valverde, S., & Solé, R. V. (2013). Networks and the city. *Architectural Design*, 83(4), 112–119.
49. Weitz, J. S., et al. (2013). Phage–bacteria infection networks. *Trends in Microbiology*, 21(2), 82–91.

2014

50. Gautrais, J., et al. (2014). The role of colony size on tunnel branching morphogenesis in ant nests. *PLOS ONE*, 9(10), e109436.
51. Hillung, J., et al. (2014). Experimental evolution of an emerging plant virus in host genotypes that differ in their susceptibility to infection. *Evolution*, 68(9), 2467–2480.
52. Solé, R. V., et al. (2014). Can a minimal replicating construct be identified as the embodiment of cancer? *BioEssays*, 36(5), 503–512.
53. Valverde, S. (2014). Evolution of patent citation networks. In *Complexity in Engineering (COMPENG)* (pp. 1–5). IEEE.

2015

54. Valverde, S., & Solé, R. V. (2015). A cultural diffusion model for the rise and fall of programming languages. *Human Biology*, 87(3), 224–234.
55. Valverde, S., & Solé, R. V. (2015). Punctuated equilibrium in the large-scale evolution of programming languages. *Journal of the Royal Society Interface*, 12(107), 20150249.
56. Valverde, S., et al. (2015). Structural determinants of criticality in biological networks. *Frontiers in Physiology*, 6, 127.

2016

57. Duran-Nebreda, S., et al. (2016). Emergence of proto-organisms from bistable stochastic differentiation and adhesion. *Journal of the Royal Society Interface*, 13(117), 20160108.
58. Flores, C. O., et al. (2016). BiMat: A MATLAB package to facilitate the analysis of bipartite networks. *Methods in Ecology and Evolution*, 7(1), 127–132.

59. Solé, R., et al. (2016). On singularities and black holes in combination-driven models of technological innovation networks. *PLOS ONE*, 11(1), e0146180.
60. Valverde, S. (2016). Major transitions in information technology. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 371(1701), 20150450.
61. Valverde, S., & García-Ojalvo, J. (2016). Hacia una teoría unificada de la criticalidad biológica. *Investigación y Ciencia*.
62. Valverde, S., & Solé, R. V. (2016). Correction to “Punctuated equilibrium in the large-scale evolution of programming languages”. *Journal of the Royal Society Interface*, 13(117), 20160272.
63. Valverde, S., et al. (2016). The software crisis of synthetic biology. *bioRxiv*, 041640.

2017

64. Aguilar, D., et al. (2017). Computational analysis of multimorbidity between asthma, eczema and rhinitis. *PLOS ONE*, 12(6), e0179125.
65. Solé, R., et al. (2017). The paradox of constant oceanic plastic debris: Evidence for evolved microbial biodegradation? *bioRxiv*, 135582.
66. Valverde, S. (2017). Breakdown of modularity in complex networks. *Frontiers in Physiology*, 8, 497.
67. Valverde, S. (2017). Visualizing the evolution of programming languages. *Leonardo*, 50(5), 505.
68. Valverde, S., et al. (2017). Spatially induced nestedness in a neutral model of phage-bacteria networks. *Virus Evolution*, 3(2), vex021.

2018

69. Corominas-Murtra, B., et al. (2018). Chromatic transitions in the emergence of syntax networks. *Royal Society Open Science*, 5(12), 181286.
70. Rosas-Casals, M., et al. (2018). A simple spatiotemporal evolution model of a transmission power grid. *IEEE Systems Journal*, 12(4), 3747–3754.
71. Valverde, S., et al. (2018). The architecture of mutualistic networks as an evolutionary spandrel. *Nature Ecology & Evolution*, 2(1), 94–99.

2020

72. Ferrando-Bernal, M., et al. (2020). Mapping co-ancestry connections between the genome of a Medieval individual and modern Europeans. *Scientific Reports*, 10(1), 6843.
73. Solé, R. V., & Valverde, S. (2020). Evolving complexity: How tinkering shapes cells, software and ecological networks. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 375(1796), 20190325.
74. Valverde, S. (2020). The long and winding road: Accidents and tinkering in software standardization. *Mètode: Revista de Difusió de la Investigació*, 2(105), 68–75.
75. Valverde, S., et al. (2020). Coexistence of nestedness and modularity in host–pathogen infection networks. *Nature Ecology & Evolution*, 4(4), 568–577.

2021

76. Bentley, R. A., et al. (2021). Neutral models are a tool, not a syndrome. *Nature Human Behaviour*, 5(7), 807–808.

77. Gómez-Marín, A., et al. (2021). Cognition, collective behaviours and consciousness (Challenge 3). In E. Herrera & J. A. Esteban (Coords.), *Brain, Mind & Behaviour* (CSIC Scientific Challenges: Towards 2030, White Paper 5). Consejo Superior de Investigaciones Científicas. [Publisher record](#).
78. Bovolenta, P., Manzanares, M., & Buceta, J. (Coords.). (2021). *Origins, (Co)Evolution, Diversity & Synthesis of Life* (CSIC Scientific Challenges: Towards 2030, White Paper 2). Consejo Superior de Investigaciones Científicas. Series coordinators: Jesús Marco de Lucas & M. Victoria Moreno-Arribas. Sergi Valverde contributed as a Challenge 5 coordinator; the specific chapter is listed in entry 79. [Institutional volume record](#).
79. Valverde, S., Ares, S., Balsa-Canto, E., Banga, J. R., Elena, S. F., Iranzo, J., & Manrubia, S. C. (2021). Evolutionary system biology (Challenge 5). In P. Bovolenta, M. Manzanares, & J. Buceta (Coords.), *Origins, (Co)Evolution, Diversity & Synthesis of Life* (CSIC Scientific Challenges: Towards 2030, White Paper 2). Consejo Superior de Investigaciones Científicas. [Published white paper](#).
80. Vidiella, B., et al. (2021). Habitat loss causes long extinction transients in small trophic chains. *Theoretical Ecology*, 14(4), 641–661.
81. Vidiella, B., et al. (2021). Structural entropy constrains dynamics in directed networks. Preprints, 2021010565.

2022

82. Duran-Nebreda, S., et al. (2022). Dilution of expertise in the rise and fall of collective innovation. *Humanities and Social Sciences Communications*, 9(1), 1–10.
83. Galiana, N., et al. (2022). Ecological network complexity scales with area. *Nature Ecology & Evolution*, 6(3), 307–314.
84. Toro-Delgado, E., et al. (2022). Butterfly–parasitoid–hostplant interactions in Western Palaearctic Hesperidae: A DNA barcoding reference library. *Zoological Journal of the Linnean Society*, 196(2), 757–774.
85. Vidiella, B., et al. (2022). A cultural evolutionary theory that explains both gradual and punctuated change. *Journal of the Royal Society Interface*, 19(196), 20220570.

2023

86. Bentley, R. A., et al. (2023). Is disruption decreasing, or is it accelerating? *Advances in Complex Systems*, 26(2), 2350006.
87. Duran-Nebreda, S., & Valverde, S. (2023). Composition, structure and robustness of lichen guilds. *Scientific Reports*, 13(1), 3295.
88. Duran-Nebreda, S., & Valverde, S. (2023). Rashevsky’s dream: A physico-mathematical foundation of history and culture. *Mètode Science Studies Journal*, 85–90.
89. Koppl, R., et al. (2023). *Explaining technology. Elements in Evolutionary Economics*. Cambridge University Press.

2024

90. Bentley, R. A., Horne, B., Borycz, J., Carrignon, S., Shteynberg, G., Vidiella, B., Valverde, S., & O’Brien, M. J. (2024). Cultural evolution, disinformation, and social division. *Adaptive Behavior*, 32(2), 189–203. DOI: [10.1177/10597123231186432](https://doi.org/10.1177/10597123231186432).
91. Duran-Nebreda, S., Bentley, R. A., Vidiella, B., Spiridonov, A., Eldredge, N., O’Brien, M. J., & Valverde, S. (2024). On the multiscale dynamics of punctuated evolution. *Trends in Ecology & Evolution*, 39(8), 734–744. DOI: [10.1016/j.tree.2024.05.003](https://doi.org/10.1016/j.tree.2024.05.003). [Research thread](#).

92. Duran-Nebreda, S., Vidiella, B., Spiridonov, A., Eldredge, N., O'Brien, M. J., Bentley, R. A., & Valverde, S. (2024). The many ways toward punctuated evolution. *Palaeontology*, 67(5), e12731. DOI: [10.1111/pala.12731](https://doi.org/10.1111/pala.12731).
93. O'Brien, M. J., Valverde, S., Duran-Nebreda, S., Vidiella, B., & Bentley, R. A. (2024). Punctuated equilibrium at 50: Anything there for evolutionary anthropology? Yes; definitely. *Evolutionary Anthropology: Issues, News, and Reviews*, 33(1), e22009. DOI: [10.1002/evan.22009](https://doi.org/10.1002/evan.22009).

2025

94. Duran-Nebreda, S., & Valverde, S. (2025). The natural evolution of computing. In J. J. Tehrani, J. Kendal, & R. L. Kendal (Eds.), *Oxford Handbook of Cultural Evolution* (pp. 459–473). Oxford University Press. DOI: [10.1093/oxfordhb/9780198869252.013.38](https://doi.org/10.1093/oxfordhb/9780198869252.013.38). Print edition published 2025; chapter first published online 2023.
95. Duran-Nebreda, S., Vidiella, B., Bentley, R. A., & Valverde, S. (2025). Fractal clusters and urban scaling shape spatial inequality in U.S. patenting. *npj Complexity*, 2(1), 30. DOI: [10.1038/s44260-025-00054-y](https://doi.org/10.1038/s44260-025-00054-y).
96. O'Brien, M. J., Vidiella, B., Duran-Nebreda, S., Bentley, R. A., & Valverde, S. (2025). Archaeology and the construction of artifact lineages: From culture history to phylogenetics. *Biological Theory*, 20(3), 189–211. DOI: [10.1007/s13752-025-00491-x](https://doi.org/10.1007/s13752-025-00491-x).
97. Valverde, S., Vidiella, B., Martínez-Redondo, G. I., Duran-Nebreda, S., Fernández, R., Bombarely, A., Rojas, A. M., & Bentley, R. A. (2025). Structural changes in Gene Ontology reveal modular and complex representations of biological function. *Molecular Biology and Evolution*, 42(6), msaf148. DOI: [10.1093/molbev/msaf148](https://doi.org/10.1093/molbev/msaf148).
98. Valverde, S., Vidiella, B., Spiridonov, A., & Bentley, R. A. (2025). The cultural macroevolution of arcade video games: Innovation, collaboration, and collapse. *Evolutionary Human Sciences*, 7, e30. DOI: [10.1017/ehs.2025.10015](https://doi.org/10.1017/ehs.2025.10015).
99. Vidiella, B., Duran-Nebreda, S., & Valverde, S. (2025). Networks: The visual language of complexity. In M. Stich & J. Carballido-Landeira (Eds.), *Nonlinear Dynamics for Biological Systems* (SEMA SIMAI Springer Series, Vol. 40, pp. 135–162). Springer Nature Switzerland, Cham. DOI: [10.1007/978-3-031-99044-1_6](https://doi.org/10.1007/978-3-031-99044-1_6).

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100. Valverde, S., Vidiella, B., & Duran-Nebreda, S. (2026). The evolutionary ecology of software: Constraints, innovation, and the AI disruption. In R. M. del Rio-Chanona, M. Pangallo, J. Bednar, E. D. Beinhocker, J. Kaszowska-Mojas, F. Lafond, P. Mealy, A. Pichler, & J. D. Farmer (Eds.), *The Economy as an Evolving Complex System IV* (pp. 416–444; ebook pp. 934–962). SFI Press, Santa Fe, NM, USA. DOI: [10.37911/9781947864696.28](https://doi.org/10.37911/9781947864696.28).
101. Vidiella, B., Duran-Nebreda, S., & Valverde, S. (2026). Hypergraphs in ecology link non-equilibrium dynamics and conservation. *Communications Physics*, 9(1), 198. DOI: [10.1038/s42005-026-02707-3](https://doi.org/10.1038/s42005-026-02707-3).
102. Fernández, R., Valverde, S., Bombarely, A., Cases, I., Handley, S. A., & Rojas, A. M. (2026). Rethinking molecular evolution through protein language model embeddings. *Trends in Genetics*. Online ahead of print. DOI: [10.1016/j.tig.2026.05.014](https://doi.org/10.1016/j.tig.2026.05.014).
103. Vidiella, B., Nicoletti, G., Fernández, R., Spiridonov, A., Sanchez, A., Hussain, S. T., & Valverde, S. (2026). Higher-order assembly and emergent persistence in biology. *Current Opinion in Systems Biology*, Article 100608. DOI: [10.1016/j.coisb.2026.100608](https://doi.org/10.1016/j.coisb.2026.100608). Research thread.

Teaching experience

Universitat Pompeu Fabra (UPF) — undergraduate & master teaching

Bachelor's Degree in Biomedical Engineering (UPF) (2011–2019)

Courses taught:

- 22106 Mathematical Biomodelling I
- 22123 Evolutionary Algorithms
- 22107 Introduction to the University and Biomedical Engineering
- 22117 Principles of Biological Design
- 22150 Modelling Complex Diseases

Bachelor's Degree in Human Biology (UPF) (2008–2016)

Courses taught:

- 20332 Basic Sciences 1

Computational Biomedical Engineering Master (UPF) (2014–2019)

Course taught:

- 32217 Complexity Science

Web: [Course website](#)

University of Valencia — graduate teaching

Master's Degree in Bioinformatics (University of Valencia) (2013–2026)

Invited Professor — Introduction to Complex Networks: Patterns & Models

Supervision of students and researchers

I have extensive experience supervising and training early-career researchers, including doctoral, postdoctoral, master's, and undergraduate students, primarily in the areas of theoretical modelling, network analysis, and complex systems. Former trainees have successfully continued their careers in academia, industry and research institutions in Spain and abroad.

PhD thesis supervision (completed)

1. **Rosas-Casals, M.** (2009). *Topological complexity of the electricity transmission network: Implications in the sustainability paradigm.*

PhD Thesis, Universitat Politècnica de Catalunya (UPC).

Current position: Profesor Titular de Universidad, Universitat Politècnica de Catalunya (UPC), Terrassa, Spain.

2. **Corominas-Murtra, B.** (2011). *A unified approach to the emergence of complex communication.*

PhD Thesis, Universitat Pompeu Fabra (UPF).

Current position: Assistant Professor (tenure track), Institute of Biology, University of Graz; Head of the research group Statistical Physics of Living Systems (Field of Excellence “Complexity of Life in Basic Research and Innovation”), Austria.

3. **Carrignon, S.** (2019). *Content-dependent biases in social learning strategies: A multiscale approach*.

PhD Thesis, Universitat Pompeu Fabra (UPF).

Current position: Research Associate (COREX project), Molecular and Cultural Evolution Laboratory (MACE Lab), Department of Genetics, Evolution and Environment (GEE), University College London (UCL), UK. Web: [Thesis repository record](#)

Master's and Bachelor's thesis supervision

1. **Roman, M.** (2020–2021). *The ontogeny of miRNA regulatory networks: Major structural transitions in *Blattella germanica**.

Master's Thesis, Institute of Evolutionary Biology (IBE, CSIC–UPF).

Current position: PhD student (Bascompte Lab), working on quantitative genetics and computational approaches to indirect effects in multi-species, multi-trait coevolutionary models.

2. **Ortiz Sánchez, I. D.** (2020–2021). *Mode and tempo of cultural evolution in video games*. Bachelor's Thesis, Universitat Pompeu Fabra (UPF).

Supervision of postdoctoral researchers

1. **Beatriu de Pinós Postdoctoral Fellowship (BP 2019)**

Funding agency: Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Generalitat de Catalunya / EU co-funded

Reference: 2019 BP 00206

Role: Responsible Researcher (Host PI)

Postdoctoral researcher: Salvador Durán-Nebreda

Project title: *Hypergraph analysis of plant ecotype responses to climate change*

Host institution: Institute of Evolutionary Biology (IBE, CSIC–UPF)

Period: 2019–2022

2. **BiodivRestore / MPA4Sustainability (Biodiversa+)**

Funding: BiodivRestore Transnational Cofund (Biodiversa+)

Role: Principal Investigator of the Spanish node

Postdoctoral researcher: Blai Vidiella-Rocamora

Host institution: Institute of Evolutionary Biology (IBE, CSIC–UPF)

Period: 2022–2024

Scientific service: committees, societies, and event organization

Editorial roles

- Editor, Transactions of the Complex Systems Society (TOCSS). Sep 2026–present. Web: [Editorial team](#).
- Guest Editor, Entropy special issue “Information Theory in Complex Networks”. 2020. Web: [Special issue](#).

Evaluation panels

- External evaluator, European Project FOC (Forecasting Financial Crises). European Union (EU). 2011–2012.
- External evaluator, European Commission, 2004–2007.

Program committees (international conferences)

- Program Committee member, Artificial Life XIII (13th Artificial Life Conference). International Society for Artificial Life. 2012.
- Program Committee member, Artificial Life XIV (14th Artificial Life Conference). International Society for Artificial Life. 2014.
- Program Committee member, ECAL 2015 (13th European Conference on Artificial Life). International Society for Artificial Life. 2015.
- Program Committee member, Artificial Life XV (15th Artificial Life Conference). International Society for Artificial Life. 2016.

Governance/scientific societies

- Council member, Complex Systems Society (CSS). CCS 2015–CCS 2018.
- Board member, Complexitat.cat (Associació Catalana per a l'Estudi dels Sistemes Complexos). 2020–2024.

Workshops/satellite meetings

- Organizer, Workshop on Complexity in Energy Infrastructures: Models, Metrics, Metaphors (ECCS 2011 Satellite Meeting). 15 Sep 2011.
- Organizer, Workshop on the Cultural Evolution of Technology: Evidence, Hypothesis and Theory (CCS 2015 Satellite Meeting). 30 Sep 2015. Web: [Satellite meeting](#)
- Co-organizer, “Evolution of Cultural Complexity” (CCS 2017 Satellite Meeting), Cancún, Mexico. 21 Sep 2017. Web: [Satellite meeting](#).

Talks, presentations and scientific outreach

Selected talks, conference presentations, seminars, workshops and scientific outreach activities.

1. (2000). Complexity in computer networks. The Global Internet: Measurement, Modeling & Analysis. Leiden, Netherlands.
2. (2004). Origins of software complexity. Software Complexity Workshop Conference (IST2004). The Hague, Netherlands.
3. (2004). Universal properties of bipartite software graphs. Workshop on Software and Complex Systems, 9th IEEE International Conference on Engineering of Complex Computer Systems. Florence, Italy.
4. (2004). Network motifs in computational networks: A case study in software architecture. Nordic Workshop on Networks. Copenhagen, Denmark.
5. (2005). Information theory of class graphs: On evolution and architectural constraints. Workshop on Complexity and Self-Properties. Brussels, Belgium.
6. (2005). Xarxes tecnològiques: del software a Internet. Curs “Evolució de Xarxes Complexes”. Barcelona, Spain.
7. (2005). Noise and reliable computation: Von Neumann revisited. Workshop on Evolution, Information and Computation: From Living to Non-Living Matter. Venice, Italy.
8. (2007). Complejidad tecnológica. Maratón Científico: Sistemas Complejos, Virus, Hormigas y Cerebros. Madrid, Spain.
9. (2008). Hierarchy and emergence in open-source networks. DELIS Final Workshop and Review Meeting. Barcelona, Spain.

10. (2008). Into the Turing's nest. Selection, Tinkering and Emergence in Complex Networks (Santa Fe Business Meeting). Googleplex, California, USA.
11. (2009). Complex networks: The architecture of life. Festival Punto y Raya (2nd edition). Barcelona, Spain.
12. (2009). Evolution of technology innovation networks. "Changing Cultures: Cultures of Change" (ATACD Project). Barcelona, Spain.
13. (2009). Percolation and modularity in insect nest networks. Workshop on Principles of Innovation in Swarm-made Architectures. Venice, Italy.
14. (26 Jan 2011). Community structure in a patient-phenotype network. MeDALL Kick-off Meeting. Barcelona, Spain.
15. (2 May 2011). Networks of innovation: From patents to open-source software. I Jornada Complexitat.CAT. Barcelona, Spain.
16. (15 Sep 2011). Hierarchy vs. self-organization in open-source networks. European Conference on Complex Systems (ECCS 2011). Vienna, Austria.
17. (5 Oct 2011). Hierarchy vs. self-organization in open-source networks. Spanish node meeting (FuturICT). Barcelona, Spain.
18. (17 May 2012–19 May 2012). SFI Workshop: Power Grids as Complex Networks. Santa Fe, New Mexico, USA.
19. (6 Jun 2012). Hierarchy vs. self-organisation in complex networks. Emergent Technologies & Design (Architectural Association, AA). London, UK.
20. (29 Nov 2012). Networks of innovation. BioDesign Research Lab Lectures. Barcelona, Spain.
21. (16 Sep 2013–20 Sep 2013). Evolution of patent citation networks. European Conference on Complex Systems (ECCS 2013). Barcelona, Spain. Web: [Event page](#)
22. (12 Jun 2014). Networks in action: Communication, computation and evolution. UPF-DTIC Research Seminars. Barcelona, Spain. Web: [Event page](#)
23. (16 Jun 2014–17 Jun 2014). Evolution of patent citation networks. Workshop on Complexity in Engineering (COMPENG 2014). Barcelona, Spain. Web: [Event page](#)
24. (4 Jul 2014). Phage-bacteria infection networks: From nestedness to modularity. Workshop on Virus Dynamics and Evolution. Barcelona, Spain. Web: [Event page](#)
25. (2014). Innovation as a complex system: Technological evolution and emergence of computational creativity. Interactivos?14 (Medialab-Prado). Madrid, Spain. Web: [Event page](#)
26. (9 Dec 2014). Major transitions in technological evolution. Workshop: Major Transitions in Natural, Synthetic and Artificial Evolution. Santa Fe Institute, USA. Web: [Event page](#)
27. (6 Mar 2015). Modelling the evolution of programming languages: Network methods in cultural evolution. Complexity and Socio-Ecological Dynamics Research Group (UPF, Dept. of Humanities). Barcelona, Spain. Web: [Event page](#)
28. (12 Apr 2015). Complex networks: The architecture of life. II Jornades Breyting. Vidreres, Spain. Web: [Event page](#)
29. (1 Jun 2015). Visualising the evolution of programming languages. 6th Leonardo Satellite Symposium at NetSci 2015. Zaragoza, Spain. Web: [Event page](#)
30. (8 Oct 2015). Leyes universales de los ecosistemas complejos. V Jornada UX/UOC (#juxuoc). Barcelona, Spain. Web: [Event page](#)
31. (15 Oct 2015). Mapping the space of programming languages. Innovation as Search on a Landscape of Possibilities. Santa Fe Institute, USA. Web: [Event page](#)

32. (30 May 2016). Major transitions in information technology. NetSci 2016. Seoul, South Korea.
33. (19 Apr 2017). Major transitions in information technology. Tempe, Arizona, USA.
Web: [Event page](#)
34. (17 Sep 2017–22 Sep 2017). Evolution of mutualistic networks by speciation-divergence dynamics. Conference on Complex Systems (CCS 2017). Cancún, Mexico. Conference dates; individual presentation date not established. Web: [CCS 2017 announcement](#).
35. (18 Dec 2018). 2001: Una odisea en el espacio. Una mirada desde la ciencia actual. Round-table participant, CCCB (Centre de Cultura Contemporània de Barcelona). Barcelona, Spain. Web: [Event](#).
36. (28 Jan 2018–2 Feb 2018). Major transitions in information technology. Workshop: Genetic Improvement of Software (Dagstuhl Seminar 18052). Dagstuhl, Germany. Web: [Event page](#)
37. (8 Feb 2018). Uncovering universal properties of evolutionary landscapes. Biosciences Seminar Series. University of Birmingham, UK.
38. (11 Feb 2019). The architecture of collective intelligence. Round table “Collective Intelligence” (Biennial Ciutat i Ciència). Barcelona, Spain.
Web: [Event page](#), [Podcast episode](#)
39. (6 Mar 2019). Culturomics. Workshop (Biennial Ciutat i Ciència). Barcelona, Spain.
40. (18 Mar 2019–21 Mar 2019). Architecture of mutualistic networks as spandrel. 10th International Conference on Complex Networks (CompleNet 2019). Tarragona, Spain.
Web: [Event page](#)
41. (21 May 2019). Emergence of intelligence in natural and artificial complex systems. Moderator, round table (Complexitat.cat / IEC). Barcelona, Spain.
Web: [Event page](#)
42. (25 May 2019–26 May 2019). The brain and the computer. Brainhack Networks. Burlington, Vermont, USA.
43. (24 Feb 2020). The neuroscience of videogames. Presenter (lecture by Diego Redolar Ripoll: “The Playful Brain”), CCCB lecture series “Gameplay, The Permanent Game”. Barcelona, Spain. Web: [Event page](#)
44. (26 Feb 2020). Major transitions in information technology. INET Complexity Economics Seminar, Oxford Martin School (University of Oxford). Oxford, UK. Web: [Event page](#)
45. (18 Sep 2020). Valverde, S., & Vidiella, B. Hidden dimensions: Coexistence of nestedness and modularity reveals the hypergraph nature of ecological architecture. TOPONETS 2020 Satellite @ NetSci 2020. Online event (original host city: Rome, Italy). Web: [TOPONETS 2020](#).
46. (2020). Duran-Nebreda, S., & Valverde, S. Colapso de Atari: El ataque de los clones. Charlas Culturales de Arcade Vintage. Ibi, Spain. Web: [Recording](#)
47. (21 Oct 2021). Complexitat i tecnologia: promeses i trampes del desenvolupament tecnològic. Casa de la Cultura de Girona. Girona, Spain. Web: [Event page](#)
48. (27 Oct 2021). Evolving complex networks: How tinkering shapes structure. Invited speaker, ReACT Satellite of CCS’21. Lyon, France. Web: [Event page](#)
49. (Oct 2021). The social perception of reality. Roundtable “¿Qué es la realidad?” (Palau Macaya). Barcelona, Spain. Web: [Recording](#)
50. (22 Nov 2021–24 Nov 2021). Biocreator: A distributed simulation for learning ecology and evolution. Selected workshop, EvoKE 2021 meeting. Barcelona, Spain. Web: [Event page](#)

51. (3 Dec 2021). The Ascent of Computer. Online seminar, European Centre for Living Technology (ECLT), Ca' Foscari University of Venice, Italy. Web: [Seminar announcement](#), [Recording](#).
52. (9 Jun 2022). Evolving networks: Optimization vs. tinkering. Third Workshop on Nonlinear Dynamics in Biological Systems. Madrid, Spain. Web: [Event page](#)
53. (14 Jun 2022–16 Jun 2022). The evolution of computing. SFI Workshop “The Structure of Technology”. Santa Fe Institute, New Mexico, USA.
Web: [Event page](#)
54. (7 Oct 2022). Hidden dimensions in ecology: From bipartite networks to hypergraphs. CBGP Seminars. Madrid, Spain. Web: [Event page](#)
55. (18 Feb 2023). PONG: Vida y muerte del arcade. Charlas Culturales de Arcade Vintage. Ibi, Spain. Web: [Recording](#)
56. (25 Jul 2023). Rise and fall of technological innovations. Seminar, XI GEFENOL Summer School on Statistical Physics of Complex Systems. Faculty of Physics, Universitat de Barcelona, Barcelona, Spain. Web: [Seminar programme](#).
57. (21 Oct 2023). La Evolución de los Videojuegos Arcade: Episodio I. Science-outreach talk, ArcadeCon 2023. Museo Arcade Vintage, Ibi (Alicante), Spain. Web: [ArcadeCon 2023 programme](#).
58. (4 Sep 2024). Is disruption decreasing, or is it accelerating? Contributed oral presentation, Science of Science session, Conference on Complex Systems (CCS 2024). Exeter, UK. Web: [Programme](#).
59. (18 Mar 2024). A complex system approach to investigating the dark proteome. Invited workshop talk, Dark Proteome 2.0, LifeHub meeting “Rewiring Life: The future implications of engineering biology”. La Cristalera, Miraflores de la Sierra, Spain. Web: [Programme](#).
60. (4 Apr 2024). Measuring diversity in complex systems (is not enough!). Biodiversity Seminar, Institute of Evolutionary Biology (IBE, CSIC–UPF). Barcelona, Spain.
61. (5 Sep 2024). La vida al borde del abismo: la gran mortandad del Pérmico. Evolución puntuada: cambios abruptos e innovación. Radio interview, “La Mecánica del Caracol” (EITB Radio Euskadi). Spain.
Web: [Radio interview](#)
62. (18 Sep 2024). Quemando el Mando (programa 142) – Ciencia y videojuegos. Radio/podcast interview (iVoox). Alicante. Spain. Web: [Podcast episode](#)
63. (22 Feb 2024). From Complexity to Simplicity: the Rise and Fall of Innovations. Invited talk, 2nd Meeting of the Spanish Chapter of the Complex Systems Society. Institut d’Estudis Catalans (IEC), Barcelona, Spain. Web: [Programme](#).
64. (23 May 2025). Nestedness, modularity and the architecture of ecological systems: From neutral models to hypergraphs. HESCOR Workshop on Network Analysis and Graph Databases. Cologne, Germany.
65. (Apr 2025). The evolving Gene Ontology: How network structure shapes functional annotations and computational biology. Biodiversity Seminars, Institute of Evolutionary Biology (IBE, CSIC–UPF). Barcelona, Spain.
66. (28 Jan 2025). (Invited talk) A multi-scale approach to punctuated evolution and human evolutionary synthesis. Opening Symposium of HEIRS (Hub for Human Evolution Research Synthesis). Faro, Portugal. Web: [Programme](#).
67. (14 Jul 2025). (Invited seminar) A multiscale approach to punctuated evolution in nature and society. Institute for Integrative Systems Biology (I2SysBio). Paterna (Valencia), Spain.
Web: [Event page](#)

68. (9 Oct 2025). (Invited seminar) A multiscale approach to punctuated evolution in nature and society. Theoretical and Experimental Ecology Station (SETE-CNRS). France.
69. (14 Jan 2026). (Invited talk) A multiscale approach to punctuated evolution in nature and society. Workshop on Mathematical Perspectives on Extinction Dynamics. International Centre for Theoretical Physics (ICTP). Trieste, Italy. Web: [Event page](#)
70. (22 Apr 2026). Tinkering, Ecosystems, and Hypergraph Thinking in the Evolution of Technology. Invited workshop presentation, 1st International ECOLITHIC Workshop: Towards a Radical Ecosystem View of Long-Term Technological Evolution in the Human Deep Past. University of Cologne, Cologne, Germany. Web: [ECOLITHIC](#).
71. (7 May 2026). Fractal clusters and urban scaling shape spatial inequality in U.S. patenting. Contributed lightning talk, session “AI, Generative and Scientific Networks”, International Conference on Complex Networks (CompleNet 2026). Faculty of Sciences, University of Zaragoza, Zaragoza, Spain. Web: [Conference programme](#).
72. (13 May 2026). The cultural macroevolution of arcade video games: Innovation, collaboration, and collapse. Contributed oral presentation (Standard), session “Cultural Evolution in Action”, Cultural Evolution Society Conference (CES 2026). UM6P, Rabat, Morocco; participated online. Web: [Conference](#). Supporting conference video: [Do video games evolve like species?](#).
73. (Sep 2026). Sardanyés, J., Arancibia, E., Pastor, J. C., & Valverde, S. Brain dynamics in human–machine–human interactions: A nonlinear and complex-network perspective for EEGs. Poster presentation, CuttingEEG Barcelona Garden 2026 (CuttingGardens 2026). Barcelona, Spain. Web: [Conference](#), [Poster](#), DOI: [10.5281/zenodo.22687065](https://doi.org/10.5281/zenodo.22687065).
74. (27 Sep 2026). The Evolutionary Ecology of Software: Constraints, Innovation, and the AI Disruption. Invited speaker, workshop “The Future of Research in the Age of Human–AI Collective Intelligence”, jointly held with the ACM Collective Intelligence Conference (CI 2026) and the ACM Conference on Human-AI Complementarity and Alignment (HCOMP 2026). Virginia Tech Academic Building One, Alexandria, Virginia, USA; participated online. Web: [Workshop programme](#).
75. (4 Nov 2026; forthcoming). Software as an Evolving Ecology: Tinkering, Spandrels, and the Growth of Artificial Complexity. Invited talk, “Thinking Complexity — 20 years at IXXI” (4–6 Nov 2026). ENS de Lyon, Lyon, France. Web: [Event](#), [Programme](#).

Software (research tools, educational tools, and professional development)

Gene Ontology evolution networks (code + data)

Role: Lead developer/maintainer

Associated publication: Valverde, S., et al. (2025). Structural Changes in Gene Ontology Reveal Modular and Complex Representations of Biological Function. *Molecular Biology and Evolution*.

Description: Open repository containing Gene Ontology network snapshots and analysis scripts used to study 15 years of Gene Ontology structural evolution, including curator-driven semantic restructuring events and modularization dynamics.

GitHub: [Source repository](#)

Zenodo (archived release): [Archived release](#)

Fractal patenting (code + data)

Role: Lead developer/maintainer

Associated publication: Duran-Nebreda, S., Vidiella, B., Bentley, R. A., & Valverde, S. (2025). Fractal clusters and urban scaling shape spatial inequality in U.S. patenting. *npj Complexity*.

Description: Reproducible research repository including (i) a generative model of spatial innovation dynamics, (ii) code to compute spatial inequality across scales, (iii) analytical derivation of the “inequality horizon,” and (iv) processed empirical data from geolocated U.S. patents (1905–2024).

GitHub: [Source repository](#)

Zenodo (archived release, DOI): [Archived release](#)

Arcade macroevolution (code + data)

Role: Lead developer/maintainer

Associated publication: The Cultural Macroeolution of Arcade Video Games: Innovation, Collaboration, and Collapse (*Evolutionary Human Sciences*).

Description: Full reproducibility package including curated dataset (7,205 arcade machines, traits, genre structure, and collaboration labels), data processing scripts, analysis notebooks, and figure-generation code. Dataset derived from metadata collected through the MAME project.

GitHub: [Source repository](#)

Zenodo (archived release, DOI): [Archived release](#)

Netlab

Role: Lead developer/maintainer

Type: Research software toolkit (complex networks)

Availability: Research software (internal distribution)

Description: Software suite for the analysis, modelling, and visualization of complex networks. The toolkit has been reused across multiple Complex Systems Lab publications over the last decade.

Impact/dissemination: Visualizations generated with Netlab have been featured in popular media, including The New York Times Magazine (12 Mar 2006) and a special Nature issue on Ecological Complexity (Nature, Vol. 442, No. 7100, 20 Jul 2006).

Link: [Nature editorial summary](#)

Netlab Online

Role: Developer/instructor (courseware design)

Type: Educational software (online applications)

Availability: Public (web-based)

Description: Collection of online applications developed for teaching activities and coursework on complex networks.

Link: [Courseware website](#)

BiMat

Role: Co-developer (collaboration with C. Flores, T. Poisot, J. Weitz)

Type: Open-source research software (MATLAB library)

Availability: Open-source

Description: Open-source MATLAB library for the analysis of modularity and nestedness in bipartite ecological networks.

Main features: modularity and nestedness computation; Shannon and Simpson diversity indices; random bipartite network generation via multiple null models; and network statistics utilities.

Link: [BiMat website](#)

Chimera

Role: Lead developer/research platform designer

Type: Research simulation platform (evolutionary + embodied systems)

Availability: Research software (internal distribution)

Description: Simulation platform for “programmable life systems” combining evolutionary dynamics with realistic physical embodiment. This platform facilitates the creation and evaluation of novel functions, built upon a computational model that incorporates emergent properties.

Applications: Adaptable to simulate microfluidic experiments and test hypotheses related to tumor growth and angiogenesis.

Pro Rally 2002

Organization: Ubisoft Entertainment

Role: Physics engine developer; software architecture supervisor

Type: Commercial software/videogame development (PlayStation 2, Nintendo GameCube)

Availability: Commercial release

Description: Racing video game developed for the PlayStation 2 and Nintendo GameCube consoles. My contributions included designing and implementing the physics engine and supervising the overall software architecture according to best engineering practices.

Red1 (El Periódico de Catalunya)

Organization: Ediciones Primera Plana

Role: Developer (desktop publishing + workflow integration)

Type: Professional software (desktop publishing)

Availability: Internal production tool

Description: Desktop publishing software used in the production of El Periódico de Catalunya. Automatic machine translation was incorporated in the production of the Catalan edition of the newspaper (first in Spain).